Program 3: Creating a Maven Project with Guava and Commons-IO

Create a Maven Project Using Terminal by following command:

mvn archetype: generate -DgroupId=com.example -DartifactId=MyMavenGuavaApp -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false

```
hreya@ShreyaM:~$ mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenGuavaApp -DarchetypeArtifactiId=maven-archetype-quickstart -Dinteract
veMode=false
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------< org.apache.maven:standalone-pom >-------------------
[INFO] Building Maven Stub Project (No POM) 1
[INFO] --------------------------------[ pom ]---------------------------------
[INFO]
[INFO] >>> maven-archetype-plugin:3.4.1:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO]
[INFO] <<< maven-archetype-plugin:3.4.1:generate (default-cli) < generate-sources @ standalone-pom <<<
[INFO]
[INFO]
[INFO] --- maven-archetype-plugin:3.4.1:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Batch mode
Downloading from central: https://repo.maven.apache.org/maven2/archetype-catalog.xml
Downloaded from central: https://repo.maven.apache.org/maven2/archetype-catalog.xml (18 MB at 3.7 MB/s)
[INFO] No archetype defined. Using maven-archetype-quickstart (org.apache.maven.archetypes:maven-archetype-quickstart:1.0)
[INFO] ----------------------------------------------------------------------------
[INFO] Using following parameters for creating project from Old (1.x) Archetype: maven-archetype-quickstart:1.0
[INFO] ----------------------------------------------------------------------------
[INFO] Parameter: basedir, Value: /home/shreya
[INFO] Parameter: package, Value: com.example
[INFO] Parameter: groupId, Value: com.example
[INFO] Parameter: artifactId, Value: MyMavenGuavaApp
[INFO] Parameter: packageName, Value: com.example
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] project created from Old (1.x) Archetype in dir: /home/shreya/MyMavenGuavaApp
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  15.168 s
[INFO] Finished at: 2026-03-17T22:25:40+05:30
[INFO] ------------------------------------------------------------------------
```

Now Navigate to the Project Directory using **cd MyMavenGuavaApp**

Edit the pom.xml File and add the code

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/maven-
v4_0_0.xsd">
  <modelVersion>4.0.0</modelVersion>
  <groupId>com.example</groupId>
  <artifactId>MyMavenGuavaApp</artifactId>
  <packaging>jar</packaging>
  <version>1.0-SNAPSHOT</version>
  <name>MyMavenGuavaApp</name>
  <url>http://maven.apache.org</url>
  <dependencies>
    <dependency>
      <groupId>junit</groupId>
      <artifactId>junit</artifactId>
      <version>3.8.1</version>
      <scope>test</scope>
    </dependency>

    <!-- https://mvnrepository.com/artifact/com.google.guava/guava -->
<dependency>
    <groupId>com.google.guava</groupId>
    <artifactId>guava</artifactId>
    <version>33.4.0-jre</version>
</dependency>
```

```
<!-- https://mvnrepository.com/artifact/commons-io/commons-io -->
<dependency>
    <groupId>commons-io</groupId>
    <artifactId>commons-io</artifactId>
    <version>2.18.0</version>
</dependency>

  </dependencies>

  <!-- Build: Configuring plugins and build settings -->
<build>
<plugins>
<!-- Example: Maven Compiler Plugin to compile Java code -->
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-compiler-plugin</artifactId>
<version>3.8.1</version>
<configuration>
<source>1.7</source>
<target>1.7</target>
</configuration>
</plugin>

<!-- Example: Maven Surefire Plugin to run tests -->
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-surefire-plugin</artifactId>
<version>2.22.2</version>
</plugin>


<plugin>
        <groupId>org.apache.maven.plugins</groupId>
        <artifactId>maven-jar-plugin</artifactId>
        <version>3.1.0</version>
        <configuration>
            <archive>
                <manifestEntries>
                    <Main-Class>com.example.App</Main-Class>
                </manifestEntries>
            </archive>
        </configuration>
    </plugin>

</plugins>
</build>
</project>
```

Modify App.java

Got to App.java using **cd src/main/java/com/example/** and **gedit App.java** to type the file with following code:

```
package com.example;

/**
 * Hello world!
 *
 */
// Import FileUtils class from Apache Commons IO library
// This is used to perform file operations like copying files
import org.apache.commons.io.FileUtils;

// Import File class to represent file and directory pathnames
import java.io.File;

// Import IOException to handle input-output errors
import java.io.IOException;

// Import ImmutableList from Google Guava library
// ImmutableList creates a list that cannot be modified after creation
import com.google.common.collect.ImmutableList;

// Define the main class
public class App
{
    // Main method - program execution starts from here
    public static void main( String[] args )
    {
        // Create an immutable list of strings (fruits)
        // This list cannot be changed (no add/remove allowed)
        ImmutableList<String> fruits = ImmutableList.of("Apple", "Banana", "Cherry");

        // Print the list of fruits to the console
        System.out.println(fruits);
```

```
        // Create a File object representing the source file
        File sourceFile = new File("source.txt");

        // Create a File object representing the destination file
        File destFile = new File("destination.txt");

        try {
            // Copy the contents of source.txt to destination.txt
            // Using Apache Commons IO FileUtils class
            FileUtils.copyFile(sourceFile, destFile);

            // Print success message if file is copied
            System.out.println("File copied successfully!");
        }
        catch (IOException e) {
            // If any error occurs during file copying,
            // this block will handle the exception

            // Print error message
            System.err.println("Error occurred while copying file: " + e.getMessage());
        }
    }
}
```

Build the Project using the command **mvn clean install** command

```
shreya@ShreyaM:~/devops/MyMavenGuavaApp$ gedit source.txt
shreya@ShreyaM:~/devops/MyMavenGuavaApp$ mvn compile
[INFO] Scanning for projects...
[INFO]
[INFO] -------------------< com.example:MyMavenGuavaApp >--------------------
[INFO] Building MyMavenGuavaApp 1.0-SNAPSHOT
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- maven-resources-plugin:2.6:resources (default-resources) @ MyMavenGuavaApp ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/shreya/devops/MyMavenGuavaApp/src/main/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.8.1:compile (default-compile) @ MyMavenGuavaApp ---
[INFO] Nothing to compile - all classes are up to date
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  10.440 s
[INFO] Finished at: 2026-03-17T19:47:46+05:30
[INFO] ------------------------------------------------------------------------
shreya@ShreyaM:~/devops/MyMavenGuavaApp$ mvn test
[INFO] Scanning for projects...
[INFO]
[INFO] -------------------< com.example:MyMavenGuavaApp >--------------------
[INFO] Building MyMavenGuavaApp 1.0-SNAPSHOT
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- maven-resources-plugin:2.6:resources (default-resources) @ MyMavenGuavaApp ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/shreya/devops/MyMavenGuavaApp/src/main/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.8.1:compile (default-compile) @ MyMavenGuavaApp ---
[INFO] Nothing to compile - all classes are up to date
[INFO]
[INFO] --- maven-resources-plugin:2.6:testResources (default-testResources) @ MyMavenGuavaApp ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/shreya/devops/MyMavenGuavaApp/src/test/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.8.1:testCompile (default-testCompile) @ MyMavenGuavaApp ---
[INFO] Nothing to compile - all classes are up to date
[INFO]
[INFO] --- maven-surefire-plugin:2.22.2:test (default-test) @ MyMavenGuavaApp ---
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running com.example.AppTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.015 s - in com.example.AppTest
[INFO]
[INFO] Results:
```

This compiles the project and creates a JAR in the target/ folder.

Run the Application:

If java-jar command give an error which is most probable, use the following command
mvn exec:java -Dexec.mainClass="com.example.App"

```
[INFO]
[INFO] --- maven-jar-plugin:3.1.0:jar (default-jar) @ MyMavenGuavaApp ---
[INFO] Building jar: /home/shreya/devops/MyMavenGuavaApp/target/MyMavenGuavaApp-1.0-SNAPSHOT.jar
[INFO]
[INFO] --- maven-install-plugin:2.4:install (default-install) @ MyMavenGuavaApp ---
[INFO] Installing /home/shreya/devops/MyMavenGuavaApp/target/MyMavenGuavaApp-1.0-SNAPSHOT.jar to /home/shreya/.m2/repository/com/example/MyMavenGuavaApp/1.0-SNAPSHOT/MyMavenGuavaApp-
1.0-SNAPSHOT.jar
[INFO] Installing /home/shreya/devops/MyMavenGuavaApp/pom.xml to /home/shreya/.m2/repository/com/example/MyMavenGuavaApp/1.0-SNAPSHOT/MyMavenGuavaApp-1.0-SNAPSHOT.pom
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  12.232 s
[INFO] Finished at: 2026-03-17T19:48:32+05:30
[INFO] ------------------------------------------------------------------------
```

This command gives error as

Error while copying file:Source 'source.txt' does not exist

Now, Create a Sample File to copy the content to destination file

```
Open                                   source.txt
                                       ~/devops/MyMavenGuavaApp
1 shreya
```

Now Run the command **mvn exec:java -Dexec.mainClass=”com.example.App”** again

```
[INFO]
[INFO] --- maven-jar-plugin:3.1.0:jar (default-jar) @ MyMavenGuavaApp ---
[INFO] Building jar: /home/shreya/devops/MyMavenGuavaApp/target/MyMavenGuavaApp-1.0-SNAPSHOT.jar
[INFO]
[INFO] --- maven-install-plugin:2.4:install (default-install) @ MyMavenGuavaApp ---
[INFO] Installing /home/shreya/devops/MyMavenGuavaApp/target/MyMavenGuavaApp-1.0-SNAPSHOT.jar to /home/shreya/.m2/repository/com/example/MyMavenGuavaApp/1.0-SNAPSHOT/MyMavenGuavaApp-
1.0-SNAPSHOT.jar
[INFO] Installing /home/shreya/devops/MyMavenGuavaApp/pom.xml to /home/shreya/.m2/repository/com/example/MyMavenGuavaApp/1.0-SNAPSHOT/MyMavenGuavaApp-1.0-SNAPSHOT.pom
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  12.232 s
[INFO] Finished at: 2026-03-17T19:48:32+05:30
[INFO] ------------------------------------------------------------------------
shreya@ShreyaM:~/devops/MyMavenGuavaApp$ mvn exec:java -Dexec.mainClass="com.example.App"
[INFO] Scanning for projects...
[INFO]
[INFO] --------------------< com.example:MyMavenGuavaApp >---------------------
[INFO] Building MyMavenGuavaApp 1.0-SNAPSHOT
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- exec-maven-plugin:3.6.3:java (default-cli) @ MyMavenGuavaApp ---
[Apple, Banana, Cherry]
File copied successfully!
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  3.040 s
[INFO] Finished at: 2026-03-17T19:49:22+05:30
[INFO] ------------------------------------------------------------------------
shreya@ShreyaM:~/devops/MyMavenGuavaApp$
```

The output shows file copied successfully.

We can see newly created **destination.txt** using gedit destination.txt if you are still in Project’s directory

```
Open                                   destination.txt
                                       ~/devops/MyMavenGuavaApp
1 shreya
```

File content is same as the content in source file.

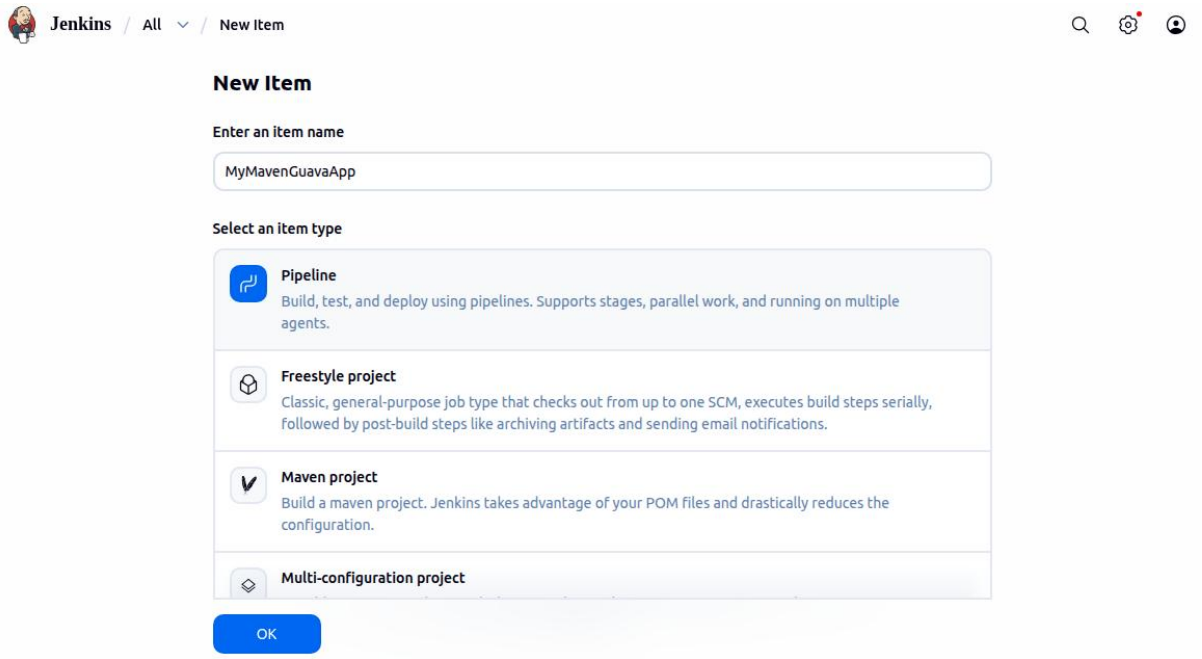
MyMavenGuava in Jenkins:

Step 1: in the terminal type the code MyMavenGuava to enter the folder .
Type the command gedit JenkinsFile to open the Jenkinsfile and paste the code

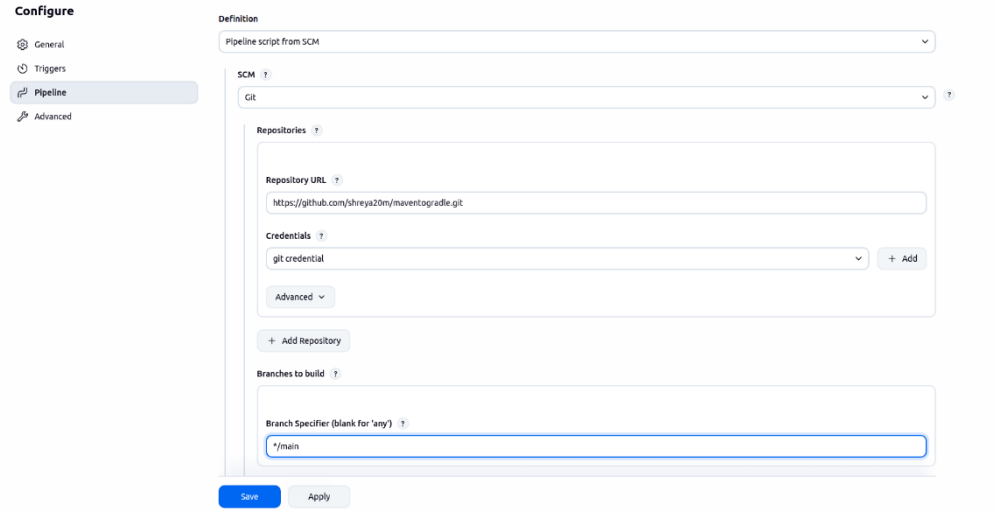
```
pipeline {
   agent any

   tools {
      maven 'Maven'
   }

   stages {
      stage('Checkout') {
         steps {
            git branch: 'master', url: 'https://github.com/shreya20m/MyMavenGuavaApp.git'
         }
      }

      stage('Build') {
```

```
            steps {
                sh 'mvn clean package -DskipTests'
            }
        }

        stage('Run Application') {
            steps {
                sh 'java -jar target/MyMavenGuavaApp-1.0-SNAPSHOT.jar'
            }
        }
    }

    post {
        success {
            echo 'Build and deployment successful!'
        }
        failure {
            echo 'Build failed!'
        }
    }
}
```

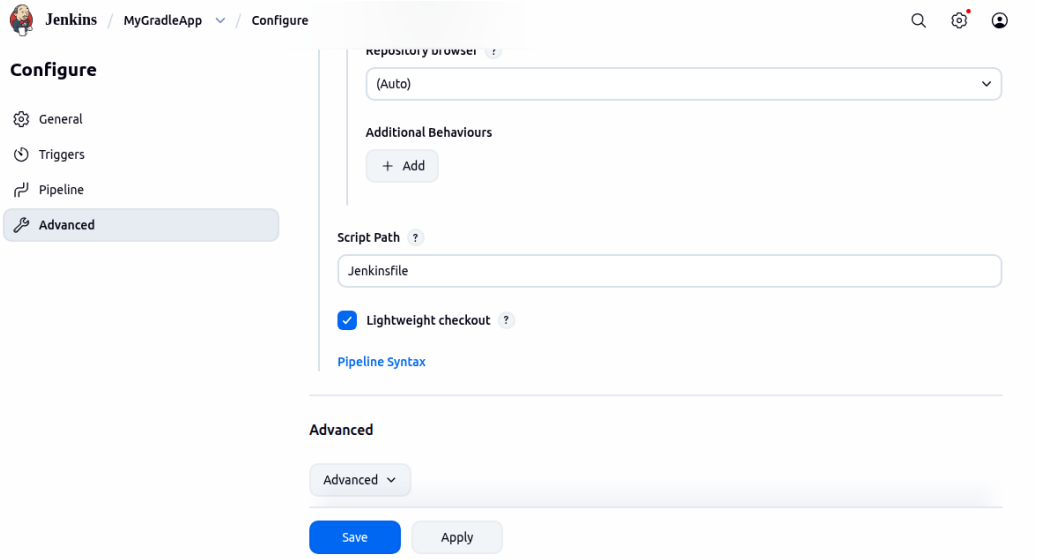
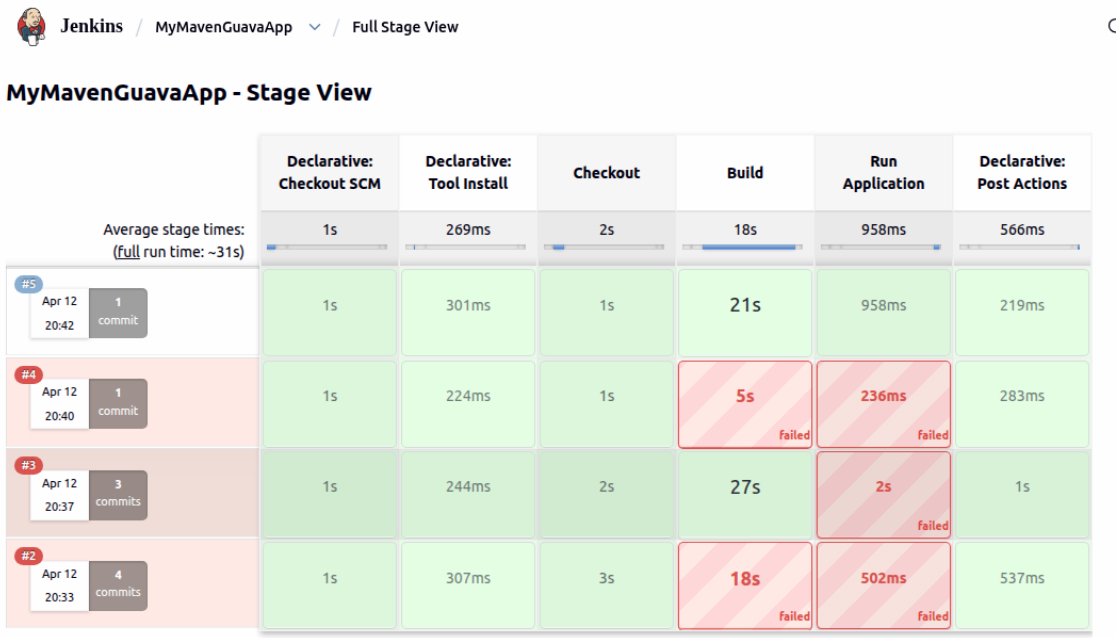
Step 2: in Jenkins Dasboard click on New Item
Give the name MyMavenGuava and select Pipeline



In configure scroll down to Pipeline and choose pipeline script from SCM,give the repository url and choose GitHub Credential



Then click on apply and save



Step 3: go to Jenkins Dasboard select the MyMavenGuava pipeline and select Build Now. After build select Full Stage View. Once the errors are corrected Build again